

ECE 50877 INTRODUCTION TO APPLIED CRYPTOGRAPHY

PROJECT DESCRIPTION & DELIVERABLES

The final project is an important part of this class. Students will work on a semester long project in groups of four. There will be multiple deliverables for projects as detailed below, with due dates announced in the Syllabus. After the project proposal, all your PDF write-up submissions must use $\LaTeX$ and follow the [USENIX format](#).

1 Project Proposal

By the project pre-proposal deadline, students should form their group and choose a relevant topic. Projects can be in any area of cryptography or broader computer security, but must be approved by the instructor. Finding good project ideas can be a daunting task. Students are encouraged to discuss ideas with the instructor in advance, and browse proceedings of security and cryptography conferences in the last few years (such as [IACR](#) workshops and conferences, [IEEE S&P](#), [USENIX Security](#), [ACM CCS](#), and [NDSS](#)). Additionally, consider and discuss other factors such as what data, compute resources, or technical knowledge is required to pursue the project.

What to submit: Each team should submit a PDF document that includes *at least two* unique project ideas. For each project idea make sure to include (i) title, (ii) problem definition, (iii) proposed methodology or solution idea, and (iv) expected findings. After the project proposal is submitted, you must schedule a meeting with the instructor to determine the final project.

You are encouraged to meet with the instructor in advance of project proposal deadline, during office hours or by appointment. You can find a project proposal sample on Brightspace.

2 Literature Review

For this project deliverable you will conduct a literature review (related work) on the determined direction. The literature review should include relevant research papers from older and more recent venues. For each paper, briefly describe what they do, if there are any shortcomings (which ideally your work plans to address), and cite them [1]. A good literature review is not just listing a bunch of papers, instead, it contextualizes the existing work and how your project idea fits within them. The related work section should be at least one page (no more than two pages) long excluding the references, and discuss at least 15 papers with 5 papers *directly* related to the project direction you are proposing with more detailed discussions.

What to submit: Each team should submit a PDF document following the [USENIX format](#) with Title, Authors, Related Work, and References sections filled in. Failure to adhere to this format will result in losing points from this deliverable grade.

3 Research Plan

By this deliverable, you have identified a problem, conducted a literature review on the topic, and should have a high level idea of your solution or approach. For this project deliverable, you report on your plan to implement and evaluate your project idea. Each team should submit a research plan including the following:

- Title (as before)
- Authors (as before)
- **Problem statement:** Clearly describe the problem (one paragraph). You should motivate the problem and specify why it is important.
- **Threat model:** Describe the security assumptions for your solution idea (several paragraphs). A threat model section should describe (i) who is the adversary, (ii) what are the goals of the adversary, (iii) what are the capabilities of the adversary, (iv) any trust assumptions made.

- **Research questions:** Include a list of at least three research questions that target your problem statement which your solution idea aims to address. Make sure that each research question is concrete and inquires about the main components of your project such as the security protection provided, performance, and scalability.
- **Proposed methodology:** A high level description of how you plan to address each research question (at least one paragraph for each). Together, this section should include details about your solution idea, how you plan to implement it, as well as evaluation process and metrics. In addition, mention any dataset, library, and computing resources you will need to perform your methodology.
- **Expected results and challenges:** Describe a set of expected outcomes by finishing this project. Also, mention any challenges you anticipate, and how you plan to address them.
- **Related work:** Literature review (as before and updated as needed).
- **References:** Your citations (as before and updated as needed).

What to submit: Each team should submit a PDF document following the [USENIX format](#) with the above sections filled in. Failure to adhere to this format will result in losing points from this deliverable grade.

4 Project Presentation

Each team should present their work to the class according to the schedule on Brightspace. All team members should present a part of the project, preferably the portion that they worked on. Each presentation should be **strictly** no more than 25 mins. Careful time management will be part of your grade. Submit a PDF of your slides. Project presentations will be graded based on the following:

- **Structure:** The presentation should motivate the problem well, briefly mention prior work, and include sufficient detail on methodology and results.
- **Organization:** The presentation and slides should be clear and easy to follow, and the talk should have an appropriate pace, adhering to the time limits. Refer to the documents posted on Brightspace (How to Give an Academic Talk and Small Guide to Giving Presentations) for tips.
- **Participation:** Your participation and engagement during presentations also counts towards a portion of your grade. Make sure to participate in Q&A of at least 2 different teams by asking the presenters questions about their project.

5 Final Report

Your report should be 8-10 pages long excluding references. Make sure to incorporate any feedback you have received on your write-ups previously. Your final report should include the following:

- Title
- Team members
- Abstract (around 200 words)
- **Introduction:** Follow the instructions in the uploaded document on “how to write the introduction”. Make sure your introduction section includes all the components specified on page 8. You can start from the problem statement previously submitted. A good introduction should be at least one page.
- **Background:** Use this section to provide any background the reader needs to know in order to understand the rest of the paper. For example, if you are building a system that uses specific cryptographic protocols, introduce and define them here.

- **Methodology:** Include your threat model and describe your methodology and implementation in detail. Start with a high level description of your approach, and then dive into details of various components. Use diagrams and figures freely. Add diagrams and figures to explain the system or approach as appropriate. app.diagrams.net is a good tool for creating high-quality diagrams. A good methodology section should be at least 2 pages.
- **Evaluation:** Present your results here and use tables or plots as appropriate. Make sure to discuss any result/table/plot you add, interpret the results, and provide explanations or justifications.
- **Discussion:** Add a section to discuss any limitation of the work (related to assumptions, methodology, practicality, etc), and directions for future work.
- **Related work:** Literature review (as before and updated as needed).
- **Conclusion:** Add a paragraph or two summarizing the insights that you obtained from working on the project, and suggestions you have on what future research should focus on. This is not a summary of your work, but the conclusions you arrived at as a result of your work.
- **References:** Your citations (as before and updated as needed).

What to submit: Each team should submit a PDF document following the [USENIX format](#) with the above sections filled in. Failure to adhere to this format will result in losing points from this deliverable grade.

References

[1] Name, “Example Citation,” Project Deliverables, 2026.